Sai Chandrasekhar Polisetty

Machine Learning & GenAI Engineer | 3+ Years Experience| +91-8465028428 |
polisettysaichandrasekhar@gmail.com | LinkedIn: https://www.linkedin.com/in/sai-chandrasekhar-polisetty-06728021b/ | GitHub: https://github.com/Saichandrasekh

Professional Summary

ML and GenAI engineer with 3.6 years of experience building LLM applications, RAG systems, chatbots, vector-search pipelines, and automated data workflows. Strong engineering background with expertise in Python, AI frameworks, and scalable backend services.

TECHNICAL SKILLS

Programming: Python, SQL

Frontend: JavaScript, React, HTML5, CSS3

Frameworks: FastAPI, Django, Flask, Rasa, LangChain

AI/ML: RAG, LLM Apps, NLP, Embeddings, Chatbots, Model Evaluation

Data: Pandas, NumPy, Matplotlib, Seaborn

Databases: MySQL, PostgreSQL (Basic), MongoDB (Basic)

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, GitHub Actions, CI/CD

Tools: Git, GitHub, Postman, Jira

Software Design: OOP, Design Patterns, Clean Code, Microservices, REST APIs

AWARDS and CERTIFICATIONS

1. Generative AI & Machine Learning Professional Certification
2. Advanced Python Full Stack Development – NxtWave CCBP 4.0

WORK EXPERIENCE

1. **Kautech IT Consulting, Hyderabad — Oct 2023 – Present**
2. **Jobshop, Bangalore — Aug 2022 – Oct 2023**

- Engineered production-ready RAG systems using transformer embeddings and FAISS, implementing retrieval evaluation (Precision@K) and prompt optimization to improve factual accuracy.
- Designed multi-turn LLM applications with structured prompting, session-based memory management, and guardrails to ensure reliable task routing and controlled responses.
- Built real-time ML pipelines for imbalanced classification, including feature engineering, model comparison, and API-based deployment using FastAPI and Docker.
- Developed a voice-enabled AI assistant integrating Whisper-based speech recognition, LLM reasoning, and neural TTS with latency and WER evaluation.
- Automated data processing workflows using Python and Pandas with validation, logging, and modular architecture to reduce manual reporting effort

Projects (GenAI & ML)

1) **Production-Grade RAG Knowledge Retrieval System**
Built a retrieval-augmented generation system using FAISS and transformer embeddings with retrieval evaluation (Precision@K) and prompt optimization to improve factual accuracy and reduce hallucinations.
2) **Real-Time Voice AI Assistant (STT + LLM + TTS)**
Designed a full-duplex voice assistant integrating Whisper-based speech recognition, context-aware LLM responses, and neural TTS with latency and WER evaluation.
3) **Credit Card Fraud Detection – End-to-End ML Pipeline**
Developed an imbalanced classification pipeline with feature engineering, SMOTE optimization, model comparison, and FastAPI-based real-time deployment.
4) **LLM Evaluation & Prompt Optimization Framework**
Built a benchmarking framework to evaluate LLM responses across prompt variants, temperature settings, and retrieval strategies, reducing hallucination rate through structured testing and logging.

EDUCATION

Bachelor of Technology (B.Tech) in Electronic and Communication
Jawaharlal Nehru Technological University, Kakinada | 2017 – 2021